

SAT Math Mastery · Week 1B**Lines, Slope & Linear Functions — Answer Key & Worked Solutions****Time** 120 minutes: 90 min solving + 30 min self-check and error log**Topics** slope from points/graphs/tables · forms of a line · intercepts · parallel & perpendicular · translations · meaning of slope/intercept · linear function notation**Name / Date** _____**1 · SKILLS & IDEAS**

- S1.** Find slope from two points, a graph, a table, a standard-form equation, or a verbal rate.
- S2.** Write and convert between slope-intercept, point-slope, standard and intercept forms.
- S3.** Find x - and y -intercepts and use them (areas, conditions, context meaning).
- S4.** Parallel and perpendicular lines, including directly from $Ax + By = C$.
- S5.** Translate a line (up/down/left/right) and track slope and intercepts.
- S6.** Interpret slope and intercept in context; build $f(x) = mx + b$ from data and solve for inputs.
- S7.** Coordinate reasoning: collinearity, midpoint, perpendicular bisector, triangles cut by axes.
- S8.** Parameter conditions on lines (a constant fixes slope, intercept, perpendicularity or area).

2 · REQUIRED FACTS

- Slope = $\frac{\text{rise}}{\text{run}}$ = constant rate of change; the y -intercept is the value at $x = 0$; the x -intercept is where $y = 0$.
- Horizontal line $y = c$: slope 0. Vertical line $x = c$: slope undefined.
- Parallel $\iff$ equal slopes (different intercepts). Perpendicular $\iff$ slopes are negative reciprocals (product -1).
- A function is linear if equal steps in x give equal steps in y (check every gap in a table).
- Shifting a graph up k adds k to y ; slope never changes under a translation.
- Two points determine a line; on a graph read only points that sit exactly on grid intersections.

3 · REQUIRED FORMULAS

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \quad y = mx + b, \quad y - y_1 = m(x - x_1), \quad \frac{x}{a} + \frac{y}{b} = 1$$

$$Ax + By = C: \quad m = -\frac{A}{B}, \quad x\text{-int } \frac{C}{A}, \quad y\text{-int } \frac{C}{B}, \quad \text{area with axes} = \frac{1}{2} \left| \frac{C}{A} \cdot \frac{C}{B} \right|$$

$$\parallel: A_1B_2 = A_2B_1, \quad \perp: A_1A_2 + B_1B_2 = 0, \quad \text{midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right), \quad f(a + d) - f(a) = md$$

4 · MINI THEORY SHEET**Choose the form by the data.**

Given	Fastest route
two points	$m = \frac{\Delta y}{\Delta x}$, then point-slope with either point
slope and y -intercept	$y = mx + b$ directly
both intercepts $(a, 0), (0, b)$	$\frac{x}{a} + \frac{y}{b} = 1$, or $m = -\frac{b}{a}$
$Ax + By = C$	read $m = -\frac{A}{B}$; set $x = 0$ or $y = 0$ for intercepts
a table	m from any two rows; confirm the rest have the same m

Perpendicular in standard form. Swap the coefficients and negate one: $\perp$ to $Ax + By = C$ is $Bx - Ay = C'$; find C' from the given point.

Parallel in standard form. Keep A, B (or scale both), change only C .

Translation. Down k : $b \rightarrow b - k$, and the x -intercept moves by $\frac{k}{m}$. Same slope, always.

Meaning. In $y = mx + b$: m = change in y per 1 unit of x ; b = starting value. Units of m are (units of y)/(units of x).

Constant conditions. Turn the condition (slope, perpendicular, area, collinear) into one equation in the constant, then solve.

Whole-number contexts. Solve $f(x) = \text{target}$; if x must be a whole day/trip/session, round to the first value that *reaches* the target.

5 · GUIDED EXAMPLES

Example 1 — Two points $\rightarrow$ equation $\rightarrow$ intercepts

A line passes through $(-1, 7)$ and $(3, -1)$. Find its equation and both intercepts.

$$m = \frac{-1 - 7}{3 - (-1)} = \frac{-8}{4} = -2. \text{ Point-slope: } y - 7 = -2(x + 1) \Rightarrow y = -2x + 5.$$

$$y\text{-intercept } 5; x\text{-intercept: } 0 = -2x + 5 \Rightarrow x = \frac{5}{2}.$$

Example 2 — Perpendicular through a point (standard form)

Line ℓ : $4x - 3y = 12$. Write the line through $(2, -1)$ perpendicular to ℓ .

Swap and negate: $3x + 4y = C'$. Plug in $(2, -1)$: $C' = 3(2) + 4(-1) = 2$. Answer: $3x + 4y = 2$.

Example 3 — Meaning of slope and intercept

A gym plan costs \$340 after 2 months and \$640 after 5 months (linear). Write $C(m)$ and interpret its parts.

$$\text{slope} = \frac{640 - 340}{5 - 2} = 100. \quad C(m) = 100m + b; \quad 340 = 200 + b \Rightarrow b = 140. \quad \text{So } C(m) = 100m + 140.$$

100 = dollars per month; 140 = joining fee (cost at $m = 0$).

6 · PRACTICE SET — 25 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 15x + 10$. What is the value of $f(x)$ when $x = 2$?

(A) 30

(B) 40 ✓

(C) 50

(D) 27

Worked solution

1. Substitute $x = 2$: $f(2) = 15(2) + 10$.

2. $f(2) = 30 + 10 = 40$.

Answer: (B)

2. EASY · NO CALCULATOR NEEDED

In the xy -plane, the graph of $y = f(x)$ is a line that passes through $(0, 21)$ and has a slope of 4. Which equation defines f ?

(A) $f(x) = -4x + 21$

(B) $f(x) = 21x + 4$

(C) $f(x) = 4x + 21$ ✓

(D) $f(x) = 4x - 21$

Worked solution

1. The line crosses the y -axis at $(0, 21)$, so $b = 21$.

2. The slope is $m = 4$.

3. $f(x) = 4x + 21$.

Answer: (C)

3. EASY · NO CALCULATOR NEEDED

A line has a slope of -3 and passes through the point $(5, -2)$. What is the y -intercept of the line?

Worked solution

1. Use $y = mx + b$ with $m = -3$ and the point $(5, -2)$: $-2 = -3(5) + b$.

2. $-2 = -15 + b$.

3. $b = 13$.

Answer: 13

4. EASY · NO CALCULATOR NEEDED

What is the x -intercept of the line given by $3x + 4y = 24$?

(A) $(8, 0)$ ✓

(B) $(0, 6)$

(C) $(0, 8)$

(D) $(6, 0)$

Worked solution

1. At an x -intercept, $y = 0$.

2. $3x = 24$, so $x = 8$.

3. The x -intercept is the point $(8, 0)$.

Answer: (A)

5.

EASY · NO CALCULATOR NEEDED

The function g is defined by $g(x) = \frac{x+1}{5}$. What is the y -intercept of the graph of $y = g(x)$ in the xy -plane?

(A) (0, 1)

(B) (−1, 0)

(C) (0, 5)

(D) $\left(0, \frac{1}{5}\right)$ ✓**Worked solution**

1. The y -intercept occurs at $x = 0$.

2. $g(0) = \frac{0+1}{5} = \frac{1}{5}$.

3. The y -intercept is $\left(0, \frac{1}{5}\right)$.

Answer: (D)

6.

EASY · NO CALCULATOR NEEDED

A line passes through the points (0, 2) and (3, 3). What is the slope of the line?

Worked solution

1. $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - 2}{3 - 0}$.

2. $m = \frac{1}{3}$.

Answer: $\frac{1}{3}$

7.

MEDIUM · NO CALCULATOR NEEDED

The function f is linear. Selected values of x and $f(x)$ are shown in the table. What is the value of $f(0)$?

x	$f(x)$
2	11
5	20

(A) 11

(B) 2

(C) 5 ✓

(D) 8

Worked solution

1. Slope: $\frac{20 - 11}{5 - 2} = 3$.

2. From $x = 2$ back to $x = 0$ is a step of -2 in x , so f drops by $3 \cdot 2 = 6$.

3. $f(0) = 11 - 6 = 5$.

Answer: (C)

8.

MEDIUM · NO CALCULATOR NEEDED

The graph of $\frac{x}{4} + \frac{y}{6} = 1$ in the xy -plane is a line. What is the slope of this line?

(A) $-\frac{2}{3}$

(B) $-\frac{3}{2}$ ✓

(C) $\frac{2}{3}$

(D) $\frac{3}{2}$

Worked solution

1. This is intercept form: intercepts $(4, 0)$ and $(0, 6)$.

2. $m = \frac{6 - 0}{0 - 4} = -\frac{3}{2}$.

Answer: (B)

9.

MEDIUM · NO CALCULATOR NEEDED

Line k is parallel to the line $y = -2x + 7$ and passes through the point $(4, 5)$. Which of the following is an equation of line k ?

(A) $y = -2x + 5$

(B) $y = \frac{1}{2}x + 3$

(C) $y = -2x + 7$

(D) $y = -2x + 13$ ✓

Worked solution

1. Parallel lines have equal slopes, so $m = -2$.

2. $5 = -2(4) + b \Rightarrow b = 13$.

3. $y = -2x + 13$.

Answer: (D)

10.

MEDIUM · NO CALCULATOR NEEDED

Line t is defined by $y = 4x + 15$. Line s is perpendicular to line t . What is the slope of line s ?

(A) $-\frac{1}{4}$ ✓

(B) $\frac{1}{4}$

(C) 4

(D) -4

Worked solution

1. Perpendicular slopes are negative reciprocals.

2. $m_s = -\frac{1}{4}$.

Answer: (A)

11.

MEDIUM · NO CALCULATOR NEEDED

A line passes through the points $(2, 11)$ and $(6, 27)$. What is an equation of the line?

(A) $y = 4x + 11$

(B) $y = 4x + 3$ ✓

(C) $y = 4x - 3$

(D) $y = 3x + 5$

Worked solution

1. $m = \frac{27 - 11}{6 - 2} = 4$.

2. Use $(2, 11)$: $11 = 4(2) + b \Rightarrow b = 3$.

3. $y = 4x + 3$. Check $(6, 27)$: $4(6) + 3 = 27$.

Answer: (B)

12.

MEDIUM · NO CALCULATOR NEEDED

A book club charges \$13 for the first book and \$11 for each additional book. Which equation gives the total cost y , in dollars, for x books, where $x \geq 1$?

(A) $y = 11x - 2$

(B) $y = 11x + 13$

(C) $y = 11x + 2$ ✓

(D) $y = 13x + 11$

Worked solution

1. Total = $13 + 11(x - 1)$ (the first book costs 13; the other $x - 1$ cost 11 each).

2. $13 + 11x - 11 = 11x + 2$.

Answer: (C)

13.

MEDIUM · NO CALCULATOR NEEDED

A linear function models the total cost, in dollars, of a field trip as $C(n) = 25n + 400$, where n is the number of students attending. What does the number 25 represent in this context?

(A) Cost of each additional student ✓

(B) Students needed to cover the fixed cost

(C) Fixed cost of the trip

(D) Cost for 25 students

Worked solution

1. In $y = mx + b$, m is the change in y per 1 unit of x .

2. Here each additional student adds \$25 to the total cost.

Answer: (A)

14.

MEDIUM · NO CALCULATOR NEEDED

For the linear function g , $g(9) = 11$ and $g(x + 1) - g(x) = 2$ for all x . Which equation defines g ?

(A) $g(x) = 2x + 9$

(B) $g(x) = 9x + 2$

(C) $g(x) = 2x + 11$

(D) $g(x) = 2x - 7$ ✓

Worked solution

1. $g(x + 1) - g(x) = 2$ means the slope is 2.

2. $11 = 2(9) + b \Rightarrow b = -7$.

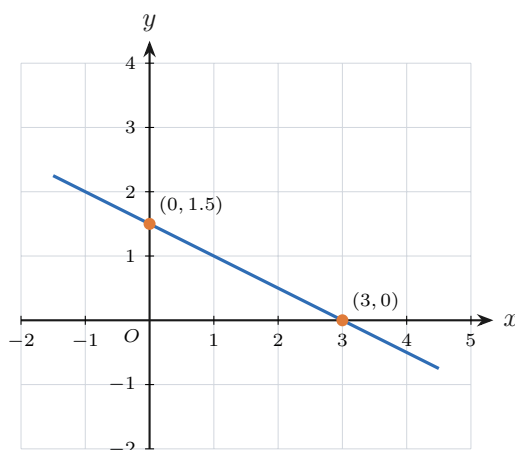
3. $g(x) = 2x - 7$.

Answer: (D)

15.

MEDIUM · NO CALCULATOR NEEDED

The graph of a linear relationship is shown. Which equation represents the line?



(A) $y = -\frac{1}{2}x + 3$

(B) $y = 2x + \frac{3}{2}$

(C) $y = -\frac{1}{2}x + \frac{3}{2}$ ✓

(D) $y = -2x + \frac{3}{2}$

Worked solution

1. Read two lattice-friendly points: (0, 1.5) and (3, 0).

2. $m = \frac{0 - 1.5}{3 - 0} = -\frac{1}{2}$; the y -intercept is 1.5.

3. $y = -\frac{1}{2}x + \frac{3}{2}$.

Answer: (C)

16.

HARD · NO CALCULATOR NEEDED

For the linear function h , $h(k) = c$ and $h(k + 5) = c + 15$, where c and k are constants. What is the value of $h(k + 2) - h(k)$?

Worked solution

1. A change of 5 in the input changes the output by 15, so the slope is $\frac{15}{5} = 3$.

2. $h(k + 2) - h(k) = 3 \cdot 2 = 6$.

Answer: 6

17.

HARD · NO CALCULATOR NEEDED

The function g is linear, and $g(a + 3) = g(a) + 15$ for all values of a . If $g(0) = -4$, what is the value of $g(6)$?

Worked solution

1. A step of 3 in the input adds 15: slope = 5.

2. $g(6) = g(0) + 5 \cdot 6 = -4 + 30 = 26$.

Answer: 26

18.

HARD · NO CALCULATOR NEEDED

If $f(x) = 4x - 3$ and $f(2a) = 3f(a) + 7$, what is the value of a ?

Worked solution

1. $f(2a) = 4(2a) - 3 = 8a - 3$.

2. $3f(a) + 7 = 3(4a - 3) + 7 = 12a - 2$.

3. $8a - 3 = 12a - 2 \Rightarrow -4a = 1 \Rightarrow a = -\frac{1}{4}$.

Answer: $-\frac{1}{4}$

19.

HARD · NO CALCULATOR NEEDED

For the linear function h , $h(3) = 20$ and $h(7) = 44$. What is the value of x for which $h(x) = 62$?

Worked solution

1. Slope: $\frac{44 - 20}{7 - 3} = 6$.

2. $h(x) = 44 + 6(x - 7)$.

3. Set equal to 62: $6(x - 7) = 18 \Rightarrow x - 7 = 3 \Rightarrow x = 10$.

Answer: 10

20.

HARD · NO CALCULATOR NEEDED

For linear function p , $p(c) = -6$, $p(5) = 42$, and the slope of the graph of p is 8. For linear function t , $t(c) = -7$ and $t(6) = 56$. What is the slope of the graph of t ?

Worked solution

1. For p : $\frac{42 - (-6)}{5 - c} = 8 \Rightarrow 5 - c = 6 \Rightarrow c = -1$.

2. For t : the points are $(-1, -7)$ and $(6, 56)$.

3. Slope of t : $\frac{56 - (-7)}{6 - (-1)} = \frac{63}{7} = 9$.

Answer: 9

21.

HARD · NO CALCULATOR NEEDED

Line ℓ passes through the point $(-2, 5)$ and is perpendicular to the line $3x - 4y = 12$. What is the y -intercept of line ℓ ?

Worked solution

1. The given line has slope $\frac{3}{4}$, so ℓ has slope $-\frac{4}{3}$.

2. $5 = -\frac{4}{3}(-2) + b = \frac{8}{3} + b$.

3. $b = 5 - \frac{8}{3} = \frac{7}{3}$.

Answer: $\frac{7}{3}$

22.

HARD · NO CALCULATOR NEEDED

$$4x - 6y = 10y + 2$$

$$ty = \frac{1}{2} + 2x$$

In the system above, t is a constant. If the system has no solution, what is the value of t ?

Worked solution

1. First equation: $4x - 16y = 2 \Rightarrow y = \frac{1}{4}x - \frac{1}{8}$.

2. Second equation: $y = \frac{2}{t}x + \frac{1}{2t}$.

3. No solution means parallel, distinct lines: $\frac{2}{t} = \frac{1}{4} \Rightarrow t = 8$.

4. Intercepts: $\frac{1}{16} \neq -\frac{1}{8}$, so the lines are distinct. $t = 8$.

Answer: 8

23.

HARD · NO CALCULATOR NEEDED

In an arithmetic sequence, the 3rd term is 17 and the 10th term is 52. What is the 20th term?

Worked solution

1. Common difference: $d = \frac{52 - 17}{10 - 3} = 5$.

2. $a_{20} = a_{10} + 10d = 52 + 50 = 102$.

Answer: 102

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, line k has slope $\frac{2}{3}$ and passes through $(6, -1)$. Line m is perpendicular to line k and has the same y -intercept as line k . What is the x -coordinate of the x -intercept of line m ?

Worked solution

1. Line k : $-1 = \frac{2}{3}(6) + b \Rightarrow b = -5$.

2. Line m has slope $-\frac{3}{2}$ and y -intercept -5 : $y = -\frac{3}{2}x - 5$.

3. Set $y = 0$: $-\frac{3}{2}x = 5 \Rightarrow x = -\frac{10}{3}$.

Answer: $-\frac{10}{3}$

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

For a positive constant K , the line $3x + 4y = K$ and the two coordinate axes bound a triangle of area 54. What is the value of K ?

Worked solution

1. Intercepts: $x = \frac{K}{3}$ and $y = \frac{K}{4}$.

2. Area = $\frac{1}{2} \cdot \frac{K}{3} \cdot \frac{K}{4} = \frac{K^2}{24}$.

3. $\frac{K^2}{24} = 54 \Rightarrow K^2 = 1296 \Rightarrow K = 36$.

Answer: 36